Yelp Restaurant Performance & Customer Engagement Analysis

Data Analytics Project · March 2026

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Project Overview

🔍 Analyze Performance

Examine restaurant success patterns using real-world Yelp data across ~52,000 establishments.

💬 Understand Engagement

Study customer behavior through reviews, tips, and check-ins to surface meaningful signals.

📈 Identify Key Factors

Uncover the metrics and patterns that distinguish top-performing restaurants from the rest.

📊 Build Dashboard

Deliver an interactive Power BI dashboard for stakeholder-ready visual insights.

About the Yelp Dataset

Dataset at a Glance

Source: Yelp Open Dataset

Scope: ~52,000 restaurants across the United States

Coverage: Multi-city, multi-state restaurant listings with rich engagement data

Key Features Included

- Restaurant name, city & state
- Star rating (1–5 scale)
- Review count & tip count
- Check-in count per location

Technology Stack Used

Python

Data cleaning and preprocessing using the **Pandas** and **NumPy** libraries for efficient data wrangling.

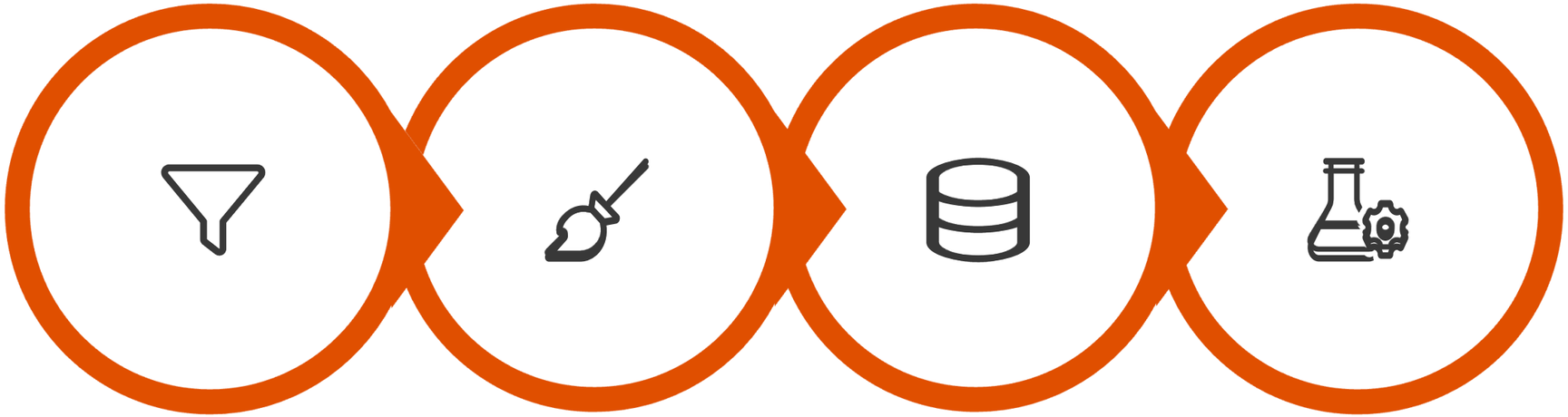
SQL / SQLite

Data aggregation, filtering, and summary table creation for structured querying of restaurant metrics.

Power BI

Interactive dashboard development for dynamic, stakeholder-ready data visualization and exploration.

Data Preparation Pipeline



Filter Clean Aggregate Engineer

Three custom metrics were engineered to power the analysis: **Rating Segment** (star-rating tiers), **Success Score** (composite of rating + engagement), and **Customer Engagement** (reviews + tips + check-ins combined).

Dashboard Overview — KPIs

52K

Restaurants

Total listings analyzed across the U.S.

3.8★

Avg Rating

Mean star rating across all restaurants

6.5M

Reviews

Total customer reviews submitted

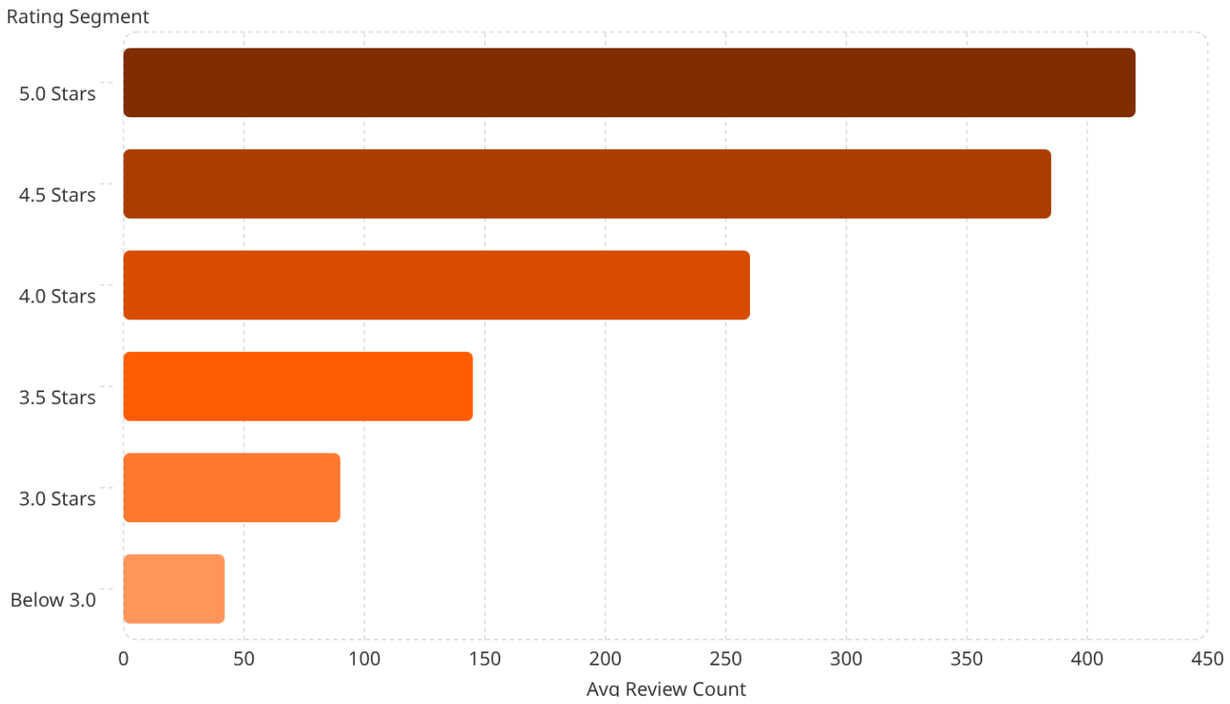
1.2M

Tips

Total tips left by Yelp users

The dashboard provides a high-level snapshot of the entire Yelp restaurant landscape, enabling quick benchmarking across markets and rating tiers.

Restaurant Performance Analysis



Key Insights

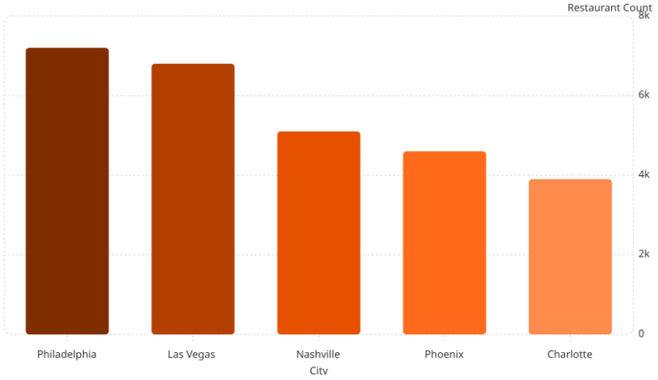
Restaurants rated **4.5+ stars** attract significantly more reviews than lower-rated peers.

A clear **positive relationship** exists between rating quality and review popularity.

The **Success Score** effectively distinguishes elite restaurants from average performers.

City Performance Analysis

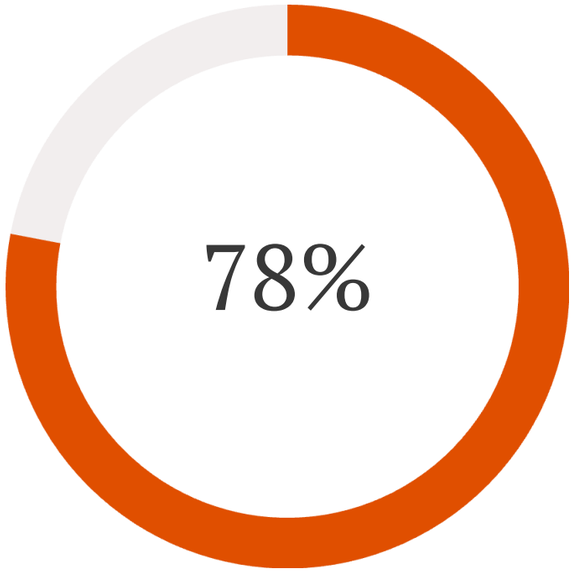
Top Cities by Restaurant Count



Geographic Insights

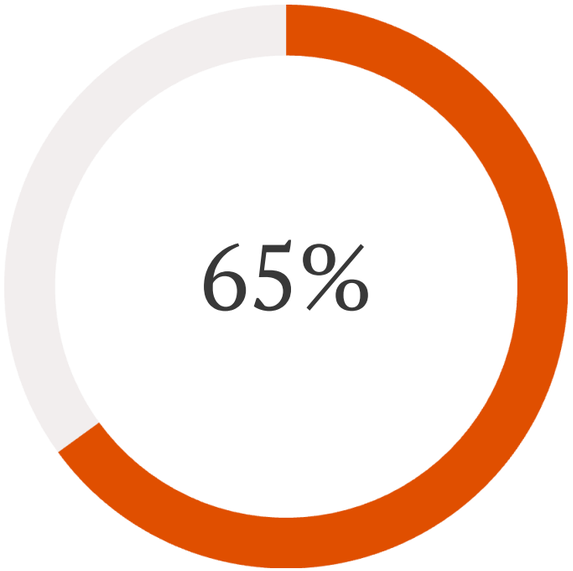
Philadelphia, Las Vegas, and Nashville lead in overall restaurant density and activity.
Cities with the highest average ratings reveal **hidden dining gems** outside major metro areas.
Restaurant distribution varies **significantly across regions**, reflecting local dining culture.

Customer Engagement Deep Dive



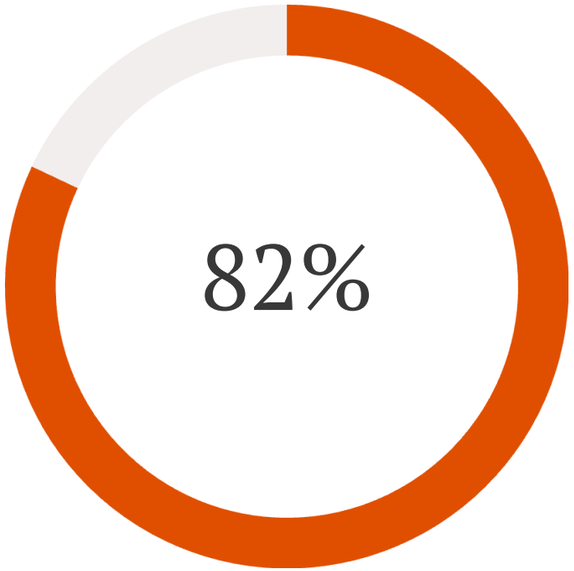
Reviews → Tips

Correlation strength between review count and tip count across all restaurants



4–5★ Share

Proportion of total engagement activity generated by top-rated restaurants



Top 3 Cities

Engagement share concentrated in Las Vegas, Philadelphia, and Nashville

Higher-rated restaurants show consistently elevated engagement across **all three metrics** — reviews, tips, and check-ins — confirming that quality and visibility are mutually reinforcing on the Yelp platform.

Key Insights & Conclusion

⭐ Rating Drives Engagement

High-rated restaurants generate significantly more reviews, tips, and check-ins.

🔗 Metrics Are Correlated

Reviews strongly co-vary with tips and check-ins — engagement is a unified signal.

🌆 City Concentration

A handful of cities dominate Yelp restaurant activity, shaping platform-wide averages.

📈 Actionable Intelligence

Engagement metrics are strong predictors of restaurant success and competitive positioning.

> By leveraging Yelp data through Python, SQL, and Power BI, this project demonstrates how data analytics can surface actionable insights that empower restaurants and stakeholders to make smarter, evidence-based decisions.